Tutorial 6

Question 1: What is opamp? Discuss the modes of operation of a typical opamp.

Question 2: What is an inverting amplifier? Determine the gain of an inverting opamp by equivalent circuit analysis.

Question 3: What is non-inverting amplifier? Determine the gain of a non-inverting opamp by equivalent circuit analysis.

Question 4: What is virtual ground? How a virtual ground is different from typical ground voltage?

Question 5: How a typical opamp can be used to sum the input voltages? Determine the output voltage of a opamp summing amplifier.

Question 6: What is an active filter? How many types of filters are there? Explain with their frequency response.

Question 7: What is active low pass filter? Design an active first order low pass filter with a cutoff frequency of 10KHz.

Question 8: What is active high pass filter? Design an active first order high pass filter with a cutoff frequency of 10KHz.

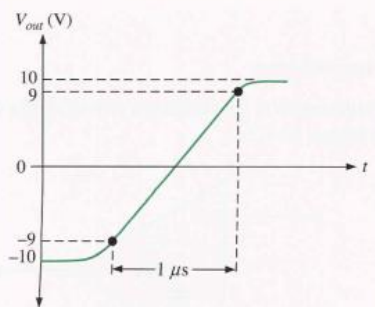
Problem 1:

A certain op-amp has an open-loop voltage gain of 100,000 and a common-mode gain of 0.2. Determine the CMRR and express it in decibels.

Problem 2:

The output voltage of a certain op-amp appears as shown in Figure 12–14 in response to a step input. Determine the slew rate.

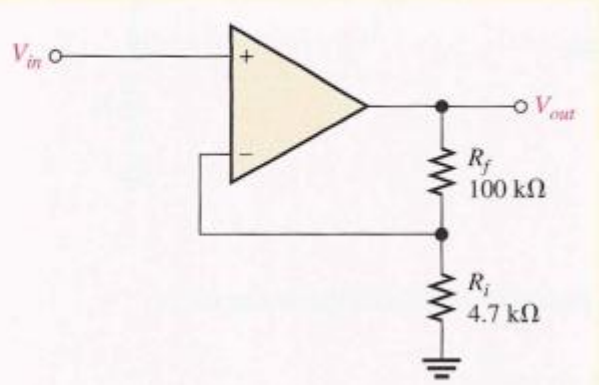
► FIGURE 12–14



Problem 3:

Determine the gain of the amplifier in Figure 12–19. The open-loop voltage gain of the op-amp is 100,000.

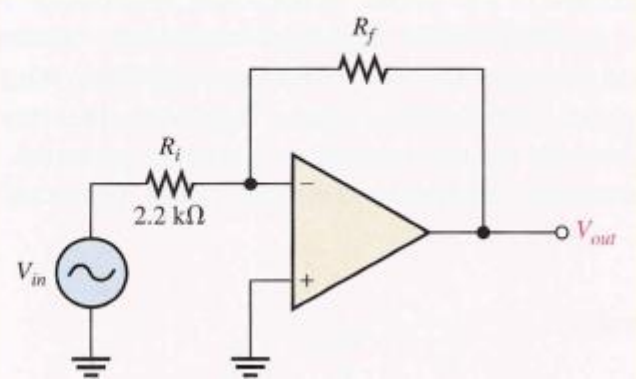
► FIGURE 12–19



Problem 4:

Given the op-amp configuration in Figure 12–23, determine the value of R_f required to produce a closed-loop voltage gain of −100.

► FIGURE 12–23

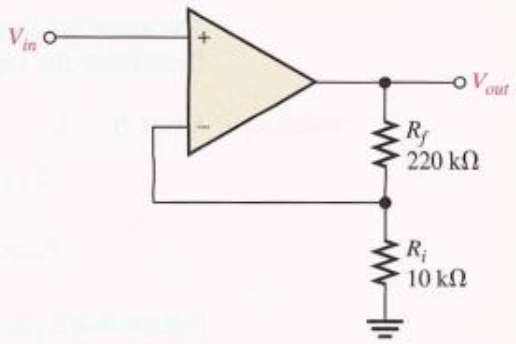


Problem 5:

(a) Determine the input and output impedances of the amplifier in Figure 12–26. The op-amp data sheet gives $Z_{in} = 2\ \text{M}\Omega$, $Z_{out} = 75\ \Omega$, and $A_{ol} = 200{,}000$.

(b) Find the closed-loop voltage gain.

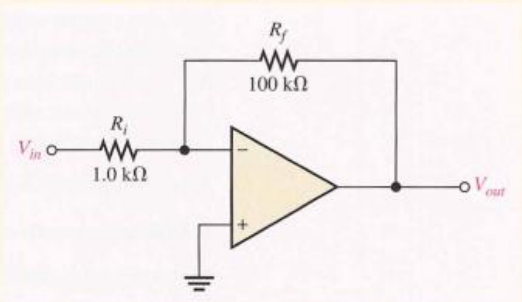
FIGURE 12–26



Problem 6:

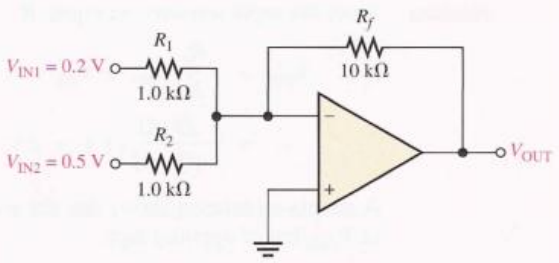
Find the values of the input and output impedances in Figure 12–29. Also, determine the closed-loop voltage gain. The op-amp has the following parameters: $A_{ol} = 50{,}000$; $Z_{in} = 4\ \text{M}\Omega$; and $Z_{out} = 50\ \Omega$.

FIGURE 12–29



Problem 7:

Determine the output voltage for the summing amplifier in Figure 13–23.



Problem 8: Find v_0 and i_0

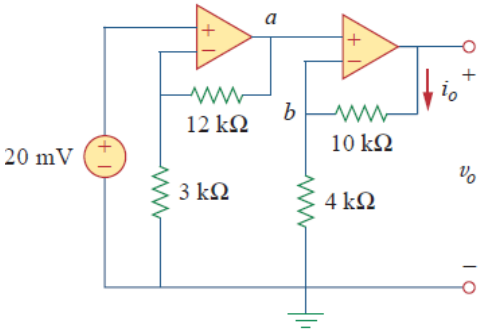
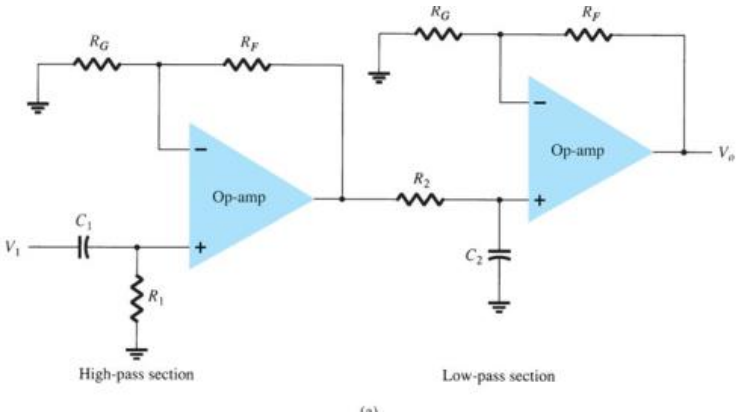


Figure 5.29

Problem 9:

EXAMPLE 11.14 Calculate the cutoff frequencies of the bandpass filter circuit of Fig. 11.34 with $R_1 = R_2 = 10\,\text{k}\Omega$, $C_1 = 0.1\,\mu\text{F}$, and $C_2 = 0.002\,\mu\text{F}$.



Problem 10:

A 741 op amp has an open-loop voltage gain of 2×10^5, input resistance of 2 MΩ, and output resistance of 50Ω. The op amp is used in the circuit of Fig. 5.6(a). Find the closed-loop gain v_o/v_s. Determine current i when $v_s = 2$ V.

